

Algae-Based Carbon Capture System: Integrated Growth Kinetics, CO₂ Sequestration, and Photobioreactor Simulation

Abstract: This study presents a computational and theoretical model of an algae-based carbon capture system integrating biological growth kinetics, CO₂ absorption dynamics, and photobioreactor design. The system leverages the Monod growth model coupled with exponential CO₂ decay to simulate carbon sequestration efficiency. The results demonstrate that algal growth rate directly influences CO₂ uptake, providing a scalable and sustainable pathway for climate mitigation.

1. Introduction: Climate change driven by rising atmospheric CO₂ levels necessitates innovative carbon capture technologies. Microalgae have emerged as a promising solution due to their high photosynthetic efficiency. This paper develops an integrated system combining growth kinetics, CO₂ capture modeling, and reactor simulation to evaluate performance.

2. Methodology: The study utilizes mathematical modeling and numerical simulation. Growth kinetics are modeled using the Monod equation, while CO₂ capture is described using a first-order decay model. A simplified photobioreactor model is incorporated to simulate real-world conditions.

3. Mathematical Models:

Monod Growth Model: $\mu = \mu_{\max} * S / (K_s + S)$

CO₂ Capture Model: $C(t) = C_0 * e^{(-kt)}$

4. Results and Discussion: The simulation shows that as substrate concentration increases, the growth rate approaches a maximum value. Concurrently, CO₂ concentration decreases exponentially over time, indicating effective carbon capture. The coupling of these models reveals that optimizing growth conditions enhances CO₂ sequestration efficiency.

5. Photobioreactor Design: A conceptual photobioreactor is proposed where algae are cultivated under controlled light, temperature, and nutrient conditions. The design ensures maximum exposure to CO₂ and efficient biomass production. Continuous flow systems further enhance capture efficiency.

6. Conclusion: The integrated model demonstrates the feasibility of algae-based carbon capture systems. By linking biological growth with CO₂ dynamics, the system provides a strong foundation for scalable environmental solutions. Future work includes experimental validation and optimization.

References:

1. Monod, J. (1949). The growth of bacterial cultures.
2. Wang, B. et al. (2008). CO₂ bio-mitigation using microalgae.